

ABHINAY MESHRAM

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SUMMARY

Machine Learning Engineer with experience building end-to-end ML solutions using Python, Scikit-learn, FastAPI, Docker, and AWS EC2. Skilled in feature engineering, model optimization with Optuna, REST API development, and deploying production-ready machine learning applications.

SKILLS

Programming Languages: Python, SQL

Machine Learning: Supervised Learning(Classification & Regression), Feature Engineering, Data Preprocessing, Model Evaluation, Cross-Validation, Hyperparameter Tuning, Ensemble Learning, Imbalanced Learning (SMOTE)

Deep Learning: Artificial Neural Networks (ANN), Convolutional Neural Network (CNN)

Data Analysis & Visualization: Exploratory Data Analysis, Statistical Analysis, Data Cleaning, Matplotlib, Seaborn

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, XGBoost, Optuna, FastAPI, TensorFlow, Keras

Deployment & APIs: REST APIs, FastAPI, Streamlit, Docker

Cloud & Tools: AWS EC2, Docker, Git, GitHub, Jupyter Notebook, Google Colab

WORK EXPERIENCE

Machine Learning Intern

Jun 2026 – Jul 2026

Cognifyz IT Solutions Pvt. Ltd. — Remote

- Developed machine learning solutions for restaurant analytics by completing projects involving restaurant rating prediction, recommendation systems, and cuisine classification.
- Performed data preprocessing, handled missing values, encoded categorical features, and prepared datasets for machine learning model development.
- Built and evaluated machine learning models for regression, classification, and content-based recommendation tasks using Python and Scikit-learn.

PROJECTS

Intel Scene Image Classification (CNN, FastAPI) [Link](#)

Aug 2026

- Built a custom **Convolutional Neural Network (CNN)** using **TensorFlow/Keras** to classify **14,034 RGB images** into six scene categories: buildings, forest, glacier, mountain, sea, and street.
- Performed image preprocessing, dataset analysis, model training, and evaluation, achieving **83.90% test accuracy** with **0.84 precision, recall, and F1-score**.
- Deployed the trained CNN as a **FastAPI REST API** for image classification, implementing image preprocessing and returning the predicted class with its confidence score.

Credit Card Fraud Detection (Python, Scikit-learn, XGBoost, FastAPI, Streamlit) [Link](#)

Jun 2026

- Built an end-to-end fraud detection system on **284,807** credit card transactions, performing EDA, feature scaling, SMOTE oversampling, and model comparison using Scikit-learn and XGBoost.
- Optimized a tuned **XGBoost** model using **RandomizedSearchCV** and **Stratified K-Fold Cross-Validation**, achieving **ROC-AUC 0.977**, **PR-AUC 0.869**, **93% precision**, and **82% recall**.
- Developed a **FastAPI REST API** and integrated it with a **Streamlit** dashboard for real-time and batch fraud prediction, probability scoring, threshold tuning, and interactive visualizations.

Customer Churn Prediction API (Python, Scikit-learn, Optuna, FastAPI, Docker, AWS EC2) [Link](#)

Jul 2026

- Built an end-to-end Machine Learning pipeline for telecom customer churn prediction using feature engineering, **ColumnTransformer**, and Scikit-learn **Pipeline**, evaluating multiple classification models including Logistic Regression, Decision Tree, KNN, Random Forest, Gradient Boosting, AdaBoost, and XGBoost.
- Optimized the final **Random Forest** model using **Optuna** for hyperparameter tuning, achieving **Accuracy 0.77**, **Precision 0.57**, **Recall 0.56**, **F1 Score 0.57**, and **ROC-AUC 0.81**.
- Developed and deployed a production-ready **FastAPI REST API**, containerized the application using **Docker**, and deployed it on **AWS EC2**, enabling real-time predictions through an interactive Swagger UI.

CERTIFICATIONS

- Machine Learning Mastery
- Data Science & Analytics with Artificial Intelligence
- Data Analysis with Python

EDUCATION

Bachelor of Science in Information Technology

University of Mumbai, Mumbai, India

Jun 2022 – Apr 2025

CGPA: 8.22 / 10